

PERIODIC TABLE

Block of the Periodic Table

s-Block

Groups 1 and 2 (except He) are called s-block elements.

They contain their valence electrons in s-sub shells.

Alkali and alkaline earth metals are present in s-block.

p-Block

Groups 13 to 18 are called p-block elements.

They contain their valence electrons in p-sub shells.

This block contains metals both metals and non-metals.

d-Block

Groups 3 to 12(or sub-group B) are called d-block elements.

They contain their valence electrons in d-sub shells.

These elements are called as transition metals.

f-Block

Two rows of elements located at the bottom of the periodic table consist of f-block elements.

They contain their valence electrons in f-sub shell.

These elements are also called lanthanides and actinides or inner transition metals.

Table 10.3: Position of Metals, Non-metals and Metalloids in the Periodic Table

1	2											11	12	13	14	15	16	17	18																																																										
H	He											B	C	N	O	F	Ne																																																												
3	4											5	6	7	8	9	10																																																												
Li	Be											Al	Si	P	S	Cl	Ar																																																												
11	12											13	14	15	16	17	18																																																												
Na	Mg											Ga	Ge	As	Se	Br	Kr																																																												
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36																																																												
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr																																																												
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54																																																												
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe																																																												
55	56	57-71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86																																																												
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn																																																												
87	88	89-103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118																																																												
Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Uut	Fl	Uup	Lv	Uus	Uuo																																																												
<table><tr><td>57</td><td>58</td><td>59</td><td>60</td><td>61</td><td>62</td><td>63</td><td>64</td><td>65</td><td>66</td><td>67</td><td>68</td><td>69</td><td>70</td><td>71</td></tr><tr><td>La</td><td>Ce</td><td>Pr</td><td>Nd</td><td>Pm</td><td>Sm</td><td>Eu</td><td>Gd</td><td>Tb</td><td>Dy</td><td>Ho</td><td>Er</td><td>Tm</td><td>Yb</td><td>Lu</td></tr><tr><td>89</td><td>90</td><td>91</td><td>92</td><td>93</td><td>94</td><td>95</td><td>96</td><td>97</td><td>98</td><td>99</td><td>100</td><td>101</td><td>102</td><td>103</td></tr><tr><td>Ac</td><td>Th</td><td>Pa</td><td>U</td><td>Np</td><td>Pu</td><td>Am</td><td>Cm</td><td>Bk</td><td>Cf</td><td>Es</td><td>Fm</td><td>Md</td><td>No</td><td>Lr</td></tr></table>																		57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
57	58	59	60	61	62	63	64	65	66	67	68	69	70	71																																																															
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu																																																															
89	90	91	92	93	94	95	96	97	98	99	100	101	102	103																																																															
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr																																																															

Metals and Non-metals

Metals:

Metals appear on the left-hand side of the periodic table, shown by grey colour.

Metals are good conductors, malleable and ductile.

Non-metals:

Non-metals are present on the right side of the periodic table, shown by yellow colour. Non-metals are bad conductors, non- malleable and non-ductile.

Metalloids:

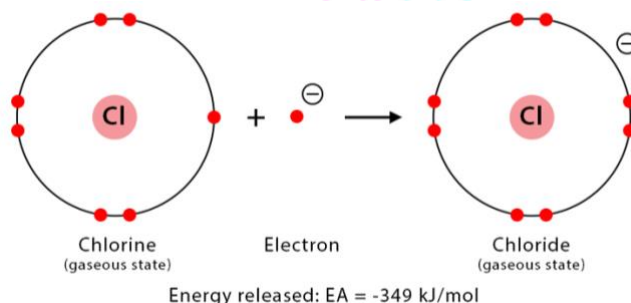
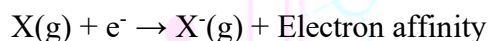
Metalloids possesses contain both the properties of metals and non-metals.

Periodicity of the properties

Electron Affinity:

Electron affinity is defined as

“The amount of energy released when an electron adds up in the valence shell of an isolated atom to form a uni-negative gaseous ion. It is also called first electron affinity.”



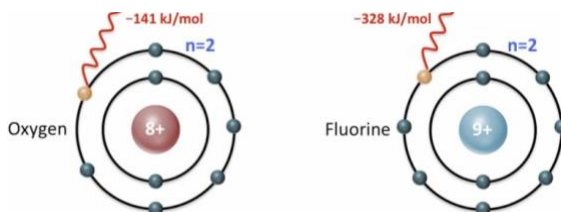
Factor affecting the electron Affinity

1. Nuclear charge:

Electron affinity tends to increase from left to right in the periodic table.

This is mainly due to the increase in nuclear charge.

As the number of protons in the nucleus increases, the attraction to electrons also increases, making it easier for the atom to accept an additional electron.

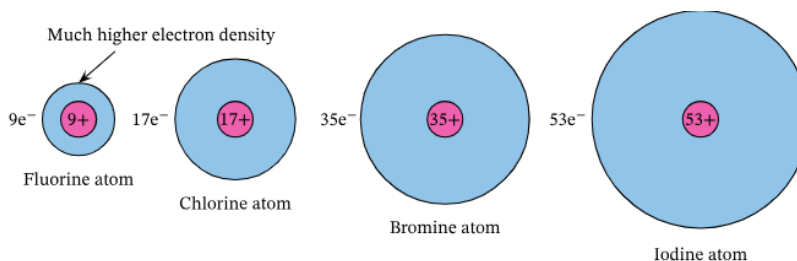


2. Atomic size:

Electron affinity generally decreases from top to bottom in a group. This is due to the increase in atomic size.

Larger atoms have electrons farther from the nucleus and the attraction between the nucleus and the outer electrons is weaker.

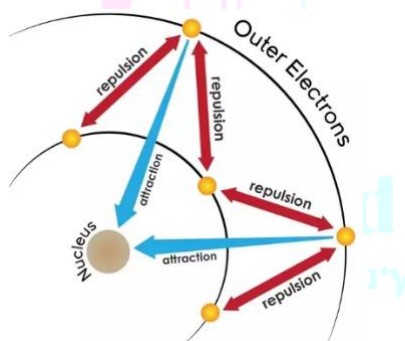
Therefore, it is more difficult for larger atoms to accept extra electron



3. Shielding effect:

The electrons in the inner shells partially shield the outer electrons from the attraction of the nucleus.

The more inner shells there are, the more the outer electrons are shielded. This shielding effect can lead to a decrease in electron affinity.



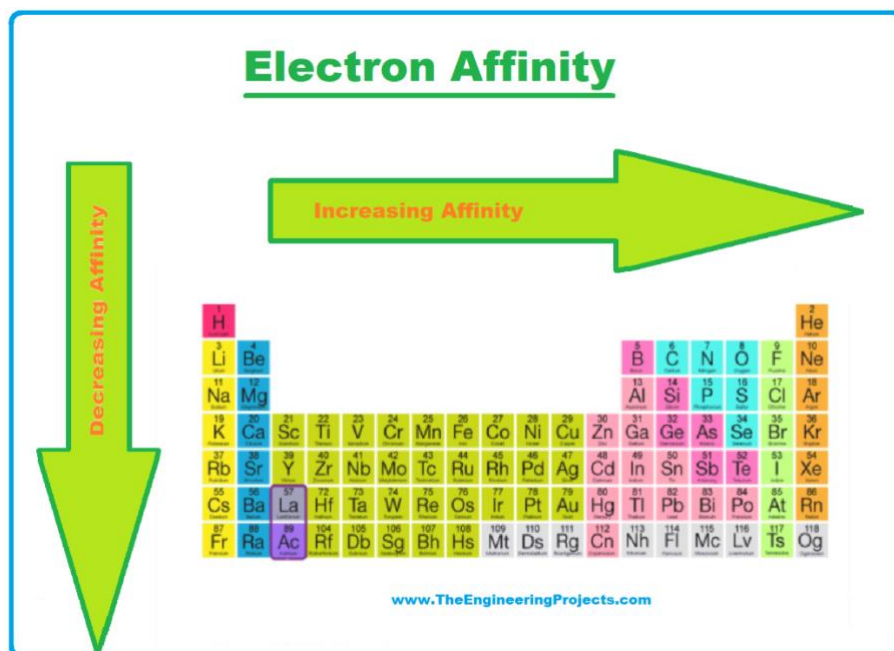
Periodic Trends

Trend in Period:

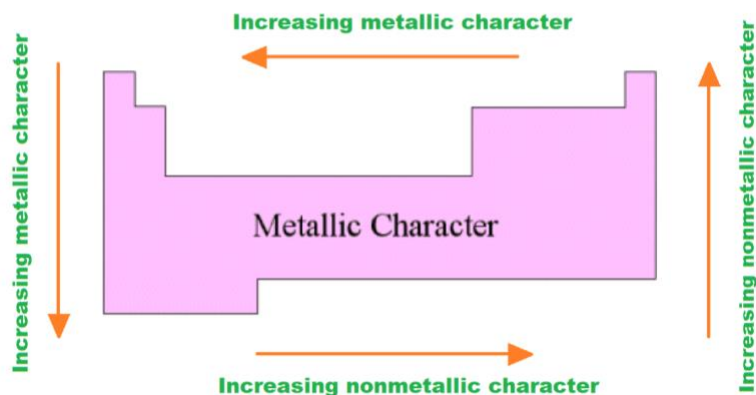
As you move from left to right in a period, electron affinity increases. This is due to an increase in nuclear charge and a decrease in atomic size, which binds the extra electron more tightly to the nucleus. The shielding effect remains the same in a period.

Trend in Group:

The electron affinity decreases from top to bottom in a group. This is due to an increase in the shielding effect. Due to increased shielding effect and increase in atomic size, the added electron binds less tightly to the nucleus. As a result, less energy is released.

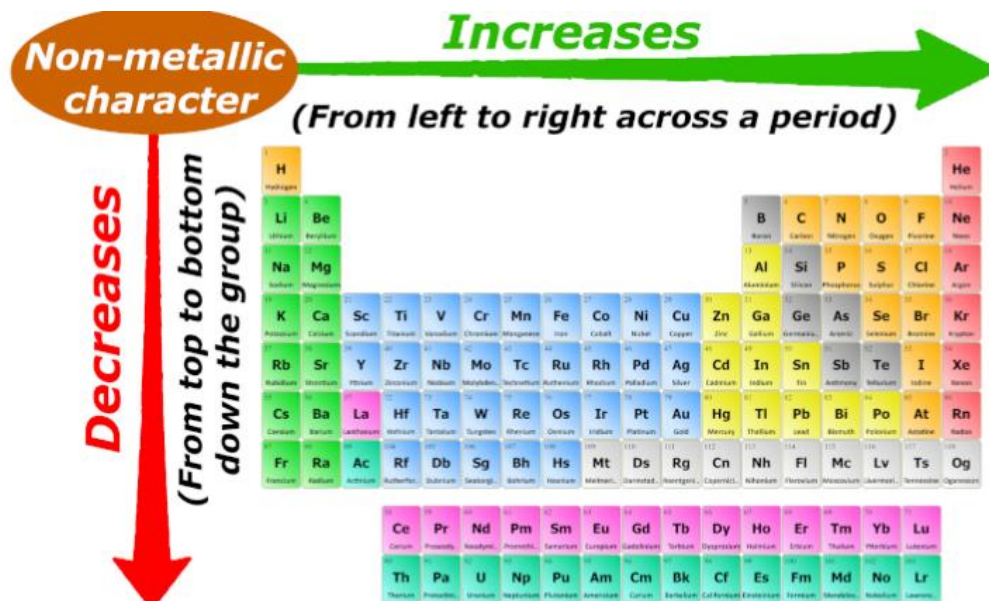
**Metallic Behaviour:**

- ✓ Metals are usually found on the left side of the periodic table.
- ✓ They tend to easily lose valence electrons and form cations to achieve stable electron configurations.
- ✓ This behavior is due to the following reasons.
- ✓ Metals usually have 1-3 valence electrons.
- ✓ Their ionization energy is usually low.
- ✓ Therefore, metals can easily lose these electrons.
- ✓ Metals have low electronegativity and do not tend to attract electrons.

Metallic Character

Non-Metallic Behaviour:

- ✓ Non-metals are usually found on the right side of the periodic table.
- ✓ Their non-metallic behavior is due to their tendency to gain or share valence electrons.
- ✓ They usually have 5-7 valence electrons.
- ✓ They have high ionization energy.
- ✓ They also have high electronegativity, which allows them to easily gain electrons or share electrons to form covalent bonds.

**Trend of metallic and non-metallic character in the group:**

- ✓ Group 1 alkali metals have a strong metallic character. Group 17 (VII A) halogens have a strong non-metallic character.
- ✓ As you move down the group, the **metallicity** of the group 1 element increases. This is due to the increase in atomic radii and their ability to easily lose electrons.
- ✓ But as we go below group 17, the non-metallic behaviour decreases. This is due to the increase in atomic radii and decrease in their ability to attract electrons.
- ✓ In group 14 (IV A), which consists of carbon (C), silicon (Si), germanium (Ge), tin (Sn), and lead (Pb), the non-metallic character decreases and the metallic character increases down the group.
- ✓ Carbon is a typical non-metal, **silicon and germanium are metalloids**, tin and lead are typical metals.

Reaction of group IA and IIA metals

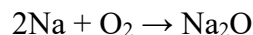
a) Reaction with oxygen:

Sodium

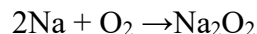
Sodium is a silvery white soft metal. It is an extremely reactive element.

It readily combines with oxygen, chlorine, and water.

In a **limited supply of oxygen**, sodium burns to form sodium oxide (Na_2O).

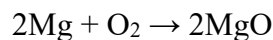


But in excess of oxygen, it forms pale yellow sodium peroxide (Na_2O_2)



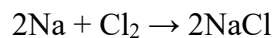
Magnesium

Magnesium is relatively hard and is also an extremely reactive element. When ignited, Mg burns in oxygen with an intense white flame to give solid magnesium oxide (MgO),

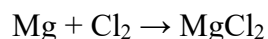


b) Reaction with chlorine

Sodium burns in chlorine with a bright yellow flame, producing a white solid, sodium chloride.

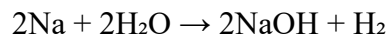


Magnesium also burns in chlorine with intense white flame, producing a white solid, magnesium chloride.



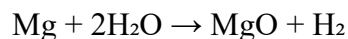
c) Reaction with water

Sodium reacts violently with cold producing hydrogen. The reaction occurs with a light explosion.



Magnesium slowly reacts with cold water to form a thin layer of magnesium hydroxide on its surface, which forms a protective coating that stops further reaction.

In steam, however, magnesium burns with a white flame, forming magnesium oxide and hydrogen.

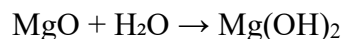


Variable oxidation state of Oxide and Halides:

Phosphorous and Sulphur show variable oxidation number because they can expand their octet e.g, in SO_2 , S has oxidation number +4, but in SO_3 , S has oxidation number +6.

Reactions of oxides with water:

Metal oxides are generally basic in character. This is because they produce bases in water ($\text{pH} > 7$).



Non-metallic oxides are generally acidic since they produce acids in water ($\text{pH} < 7$).



Trend of Acidic and Basic Character

The basic character of oxides decreases as we move along a period. This indicates decrease in metallic character. Also, the electronegativity difference between these elements and oxygen decreases as we move across the period. So, the bonding character also changes from ionic to covalent.

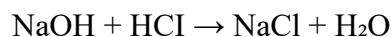
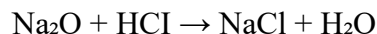
Oxides of sodium and magnesium are ionic compounds. They possess high melting and boiling points. They are soluble in water. Their aqueous solutions are good conductor of electricity.

Other oxides are covalent in nature. Oxides of P, S and Cl exist as gaseous molecules. They have low melting and boiling points.

The elements of the third period form oxides such as Na_2O , MgO , Al_2O_3 , SiO_2 , P_4O_{10} , SO_2 , Cl_2O_7 . These oxides turn from strongly basic through weakly basic, amphoteric, weakly acidic to strongly acidic.

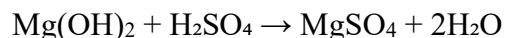
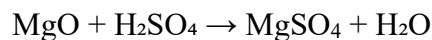
Oxide and hydroxide of sodium are strongly basic.

They react with acids to form salt and water.



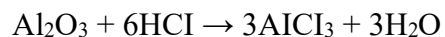
Oxide and hydroxide of magnesium are also strongly basic.

Less basic than those of sodium.



Oxide and hydroxide of aluminium are amphoteric.

They react with both acids and bases



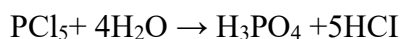
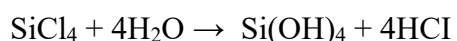
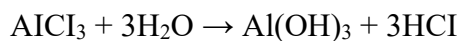
Group	IA	IIA	IIIA	IVA	VA	VIA	VIIA
3 rd period	Na	Mg	Al	Si	P	S	Cl
Oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂	Cl ₂ O ₇
Nature of oxide	Strongly basic	Basic	Amphotric (both acidic and basic)	Weakly acidic	Acidic	Strongly Acidic	Very strongly acidic

Reactions of Chlorides with water

Metal chlorides, such as sodium chloride and magnesium chloride dissolve in water forming a neutral solution.

Aluminium chloride (AlCl₃), silicon tetra chloride (SiCl₄) and phosphorous penta chloride (PCl₅) are covalent compounds.

They are also soluble in water, but they react more vigorously with water forming acidic solution. These solutions have low pH.



Type of chemical bonding in chlorides and oxides

Ionic compounds usually have higher melting points and boiling points, dissolve in water, and conduct electricity when molten.

Covalent compounds often have lower freezing and boiling points, are insoluble in water, and do not conduct electricity.

For More Information:

WhatsApp: [0306-7395469](https://www.whatsapp.com/channel/0029Vb6Hu0wEKyZO9OSkPz13)

Join my WhatsApp Channel: Follow the Sir Amjad (Chemistry Info) channel on WhatsApp:
<https://whatsapp.com/channel/0029Vb6Hu0wEKyZO9OSkPz13>

For chemistry preparation, join the following groups:

SSC-I: <https://chat.whatsapp.com/CIKJMYFmE5vFbgg7oRMOsU>

SSC-II: <https://chat.whatsapp.com/DMYmTNYItL98P7IC5cwSoj>

HSSC-I: <https://chat.whatsapp.com/DMYmTNYItL98P7IC5cwSoj>

HSSC-II: <https://chat.whatsapp.com/DMYmTNYItL98P7IC5cwSoj>

To subscribe my YouTube channel:

<https://youtube.com/@siramjadchemist?si=O56Zh-WhQIQM7mnJ>

